25 $G(r) = \dfrac{3r^{\frac{3}{2}} + r^{\frac{3}{2}}}{r}$

$= 3r^{\frac{1}{2}} + r^{\frac{3}{2}}$

$G'(r) = 3 \times \frac{1}{2} \times r^{-\frac{1}{2}} + \frac{3}{2} r^{\frac{1}{2}}$

$= \frac{3}{2} \times \frac{1}{\sqrt{r}} + \frac{3}{2}\sqrt{r}$

$v^{\frac{4}{3}}$ $\quad \frac{1}{\sqrt[3]{v^4}}$

26. $G(t) = \sqrt{5t} + \frac{\sqrt{5}}{t}$

$G'(t) = \frac{1}{2}(5t)^{-\frac{1}{2}} - \sqrt{5}\, t^{-2}$

$= \frac{1}{2} \frac{1}{\sqrt{5t}} - \frac{\sqrt{5}}{t^2}$

$\sqrt[3]{v} = v^{\frac{1}{3}}$

$v^{\frac{3}{2}} = \sqrt[2]{v^3}$

32. $f(v) = \dfrac{\sqrt[3]{v} - 2ve^x}{v}$

$= v^{-\frac{2}{3}} - 2e^x$

$\frac{1}{3} - 1$

$f'(v) = \frac{1}{2} v^{-\frac{1}{2}} - 2e^x$

$= \frac{1}{2} \frac{1}{\sqrt{v}} - 2e^x$

35. $y = tx^2 + t^3 x$

$\frac{dy}{dx} = 2tx + t^3$ $\qquad$ $\frac{dy}{dt} = x^2 + 3t^2 x$

36. $y = \frac{t}{x^2} + \frac{x}{t}$

$\frac{dy}{dx} = t(-2x^{-3}) + \frac{1}{t}$ $\qquad$ $\frac{dy}{dt} = \frac{1}{x^2} + x(-t^{-2})$

$= \frac{-2t}{x^3} + \frac{1}{t}$ $\qquad$ $= \frac{1}{x^2} - \frac{x}{t^2}$

$t \cdot (x^{-2})$

$-2tx^{-3}$

$y = \frac{3x}{1+5x^2}$ $(1, \frac{1}{2})$

$u = 3x$ $v = 1+5x^2$

$u' = 3$ $v' = 2\cdot 5x = 10x$

$y' = \frac{vu' - uv'}{v^2} = \frac{(1+5x^2)3 - 3x\cdot 10x}{(1+5x^2)^2}$

$= \frac{3+15x^2-30x^2}{1+10x^2+25x^4}$ 不用展开

$= \frac{3-15x^2}{1+10x^2+25x^4}$

$y'(1) = \frac{3-15\times 1}{(1+5\times 1)^2} = \frac{3-15}{36} = \frac{-12}{36} = -\frac{1}{3}$

$y - \frac{1}{2} = -\frac{1}{3}(x-1)$

$y = -\frac{1}{3}x + \frac{1}{3} + \frac{1}{2}$

$y = -\frac{1}{3}x + \frac{5}{6}$

$y = x + xe^x$, $(0,0)$

$y' = 1 + xe^x + e^x$

$y'(0) = 1 + 0 + 1 = 2$

$y - 0 = 2(x - 0)$

$y = 2x$

12. $f(x) = e^x \sin x + \cos x$

$f'(x) = e^x \cos x + e^x \sin x - \sin x$

15. $y = \dfrac{x}{2 - \tan x}$

$u = x \qquad v = 2 - \tan x$

$u' = 1 \qquad v' = 2 - \sec^2 x = 2 - \dfrac{1}{\cos^2 x}$

$$y' = \frac{vu' - uv'}{(2-\tan x)^2} = \frac{(2-\tan x) - x\left(2 - \frac{1}{\cos^2 x}\right)}{(2-\tan x)^2}$$

$\to x \cdot \sec^2 x$

$$= \frac{2 - \tan x - 2x + \frac{x}{\cos^2 x}}{(2-\tan x)^2}$$

21. $f(\theta) = \theta \cos\theta \sin\theta$

$f(\theta) = \theta \sin\theta \cos\theta$

$u = \theta \quad , \quad v = \sin\theta\cos\theta$

$u' = 1 \quad , \quad v' = -\sin^2\theta + \cos^2\theta$

$f'(\theta) = uv' + u'v$

$= \theta(-\sin^2\theta + \cos^2\theta) + \sin\theta\cos\theta$

$= -\theta\sin^2\theta + \theta\cos^2\theta + \sin\theta\cos\theta$